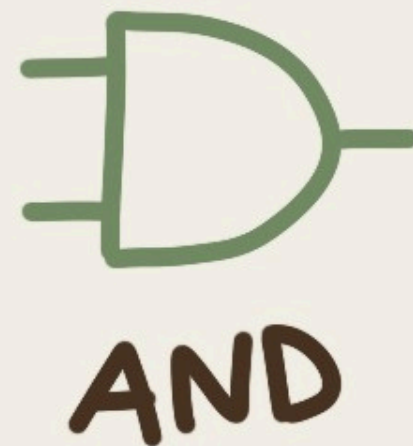


Garden Gates: Boolean Logic Glossary

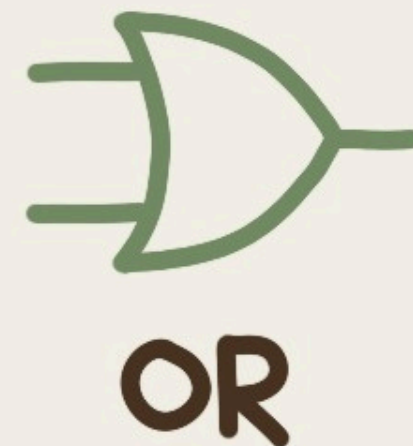
Each garden friend helps introduce a basic logic gate used in computing. Here's how their behavior maps to simple Boolean Logic:

ANDie the Ant



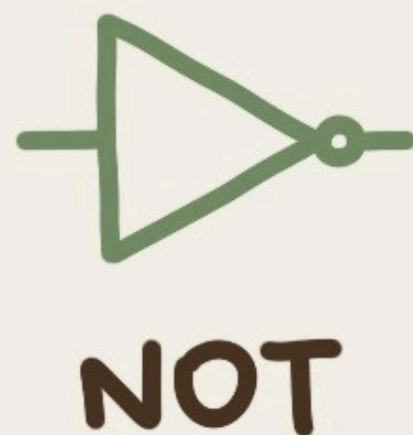
True only if both inputs are true

ORla the Owl



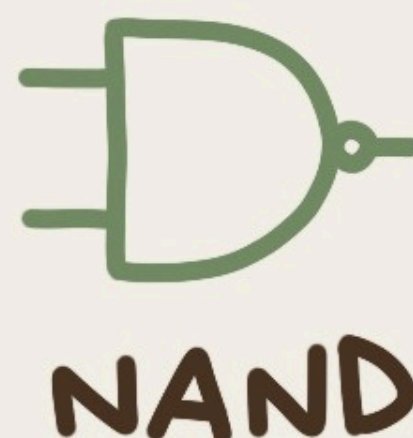
True if any input is true

NOTsy the Nightcrawler



Reverses the input: true becomes false, false becomes true

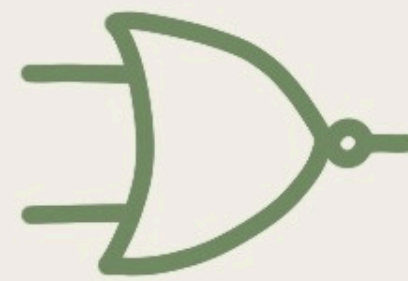
NANDi the Newt



Opposite of AND; false only if both inputs are true



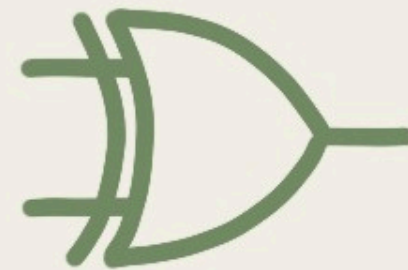
NORbert the Snail



NOR

Opposite of OR;
true only if both
inputs are false

XORa the Firefly



XOR

True if exactly one
input is true

XNORdra the
Butterfly



XNOR

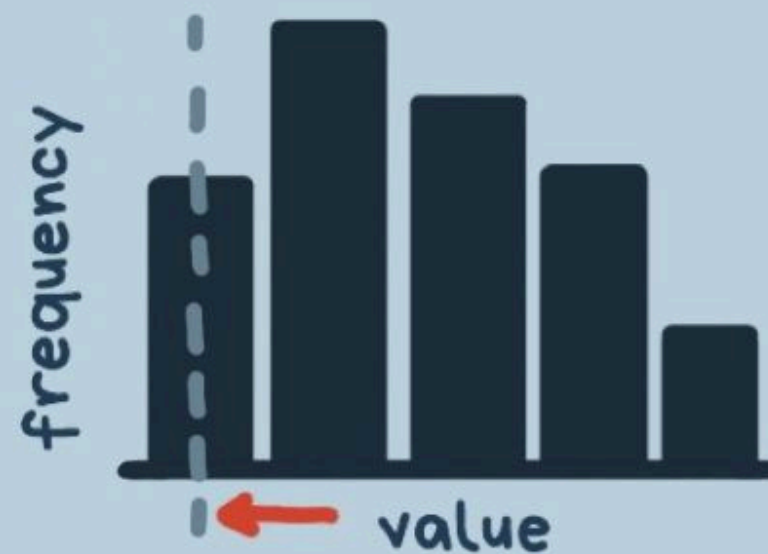
True if both inputs
are the same

Just like the garden friends work together to clean the garden,
logic gates can be combined to build complex systems—from
computers to robots to magical garden adventures!

Sea of Stats: Statistics Glossary

Each sea creature introduces a basic statistic concept used in data science. Here's how their actions reflect these ideas:

MINDy the Minnow



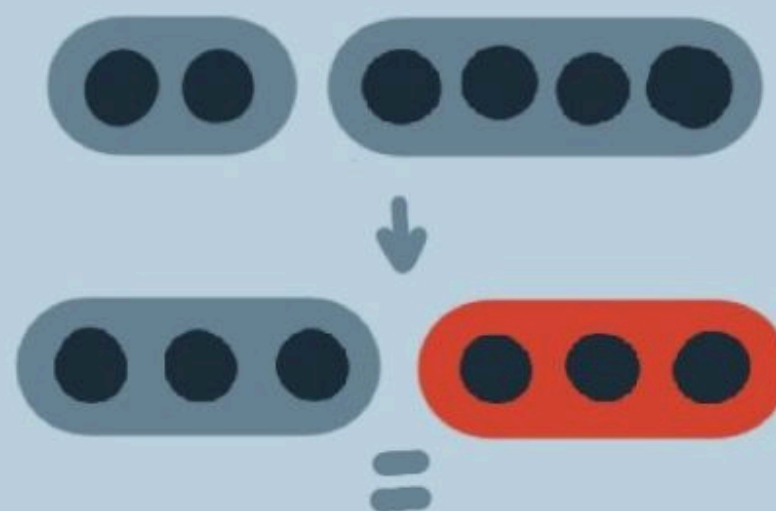
Selects the smallest value in a data set

MAXimus the Manta Ray



Selects the largest value in a data set

seMEANa the Seahorse



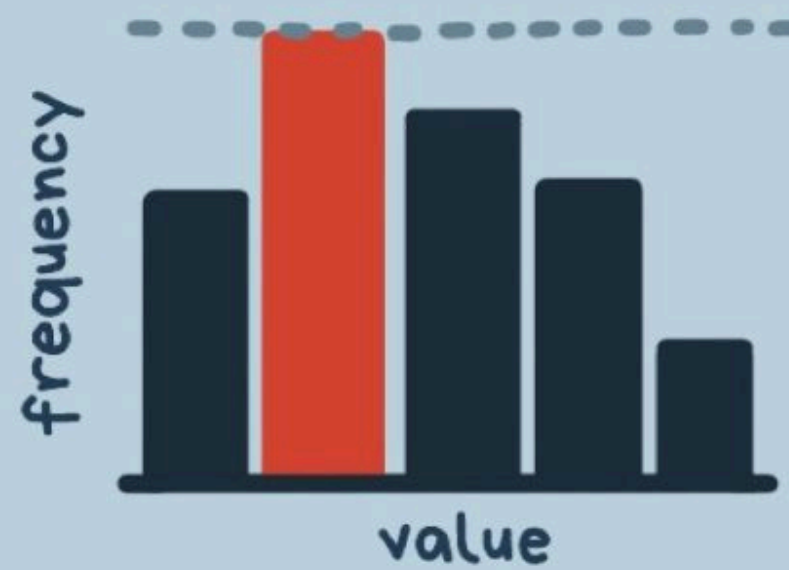
The total divided evenly among the group; the average

MEDusa the Moon Jellyfish



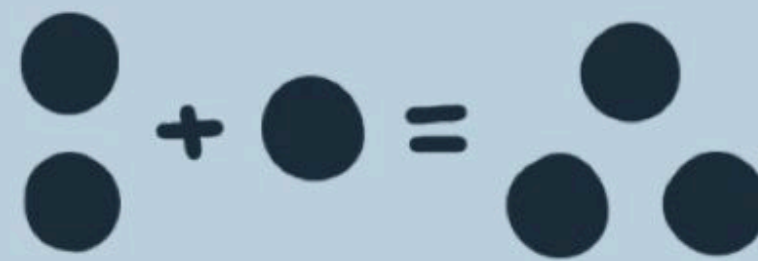
Finds the middle value in a data set

amMODE the
Angelfish



Finds the value that
appears most often

SUMMer the
Sea Turtle



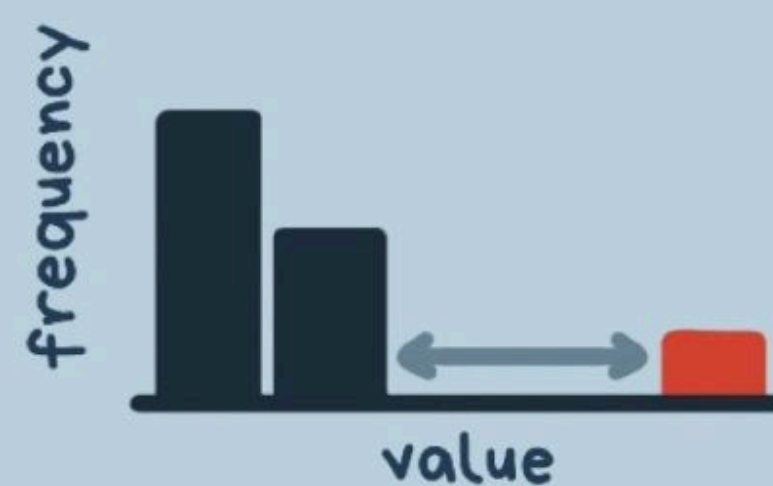
Adds all the values
in a data set

RANGella the
Reef Shark



Shows the difference
between smallest
and largest values

OUTLIERo the
Octopus



Spots values far
away from most
other values

Just like the reef friends work together to understand their busy ocean, statistics can be combined to make sense of big, messy groups—from ocean surveys to magical undersea adventures!